15.0.3 Lab - Creating Codes

Computer Networks Security

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15.0.3 Lab - Creating Codes

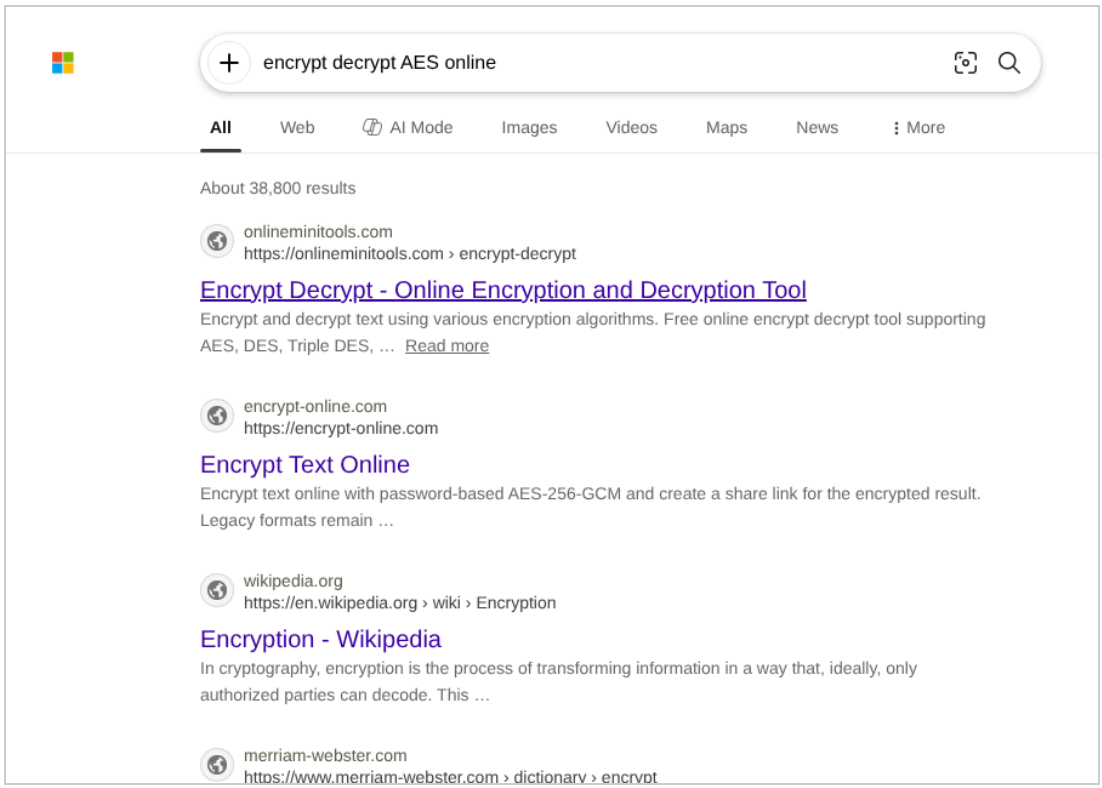
Scenario

In this lab I encrypted and decrypted messages with an online AES tool. AES (Advanced Encryption Standard) is a symmetric algorithm: the same secret key is used to encrypt and to decrypt, so both sides have to share the key in advance. The plain message is called plaintext and the encrypted message is called ciphertext. The goal is to send a message that only the person with the key can read, and to see what happens when the key is wrong.

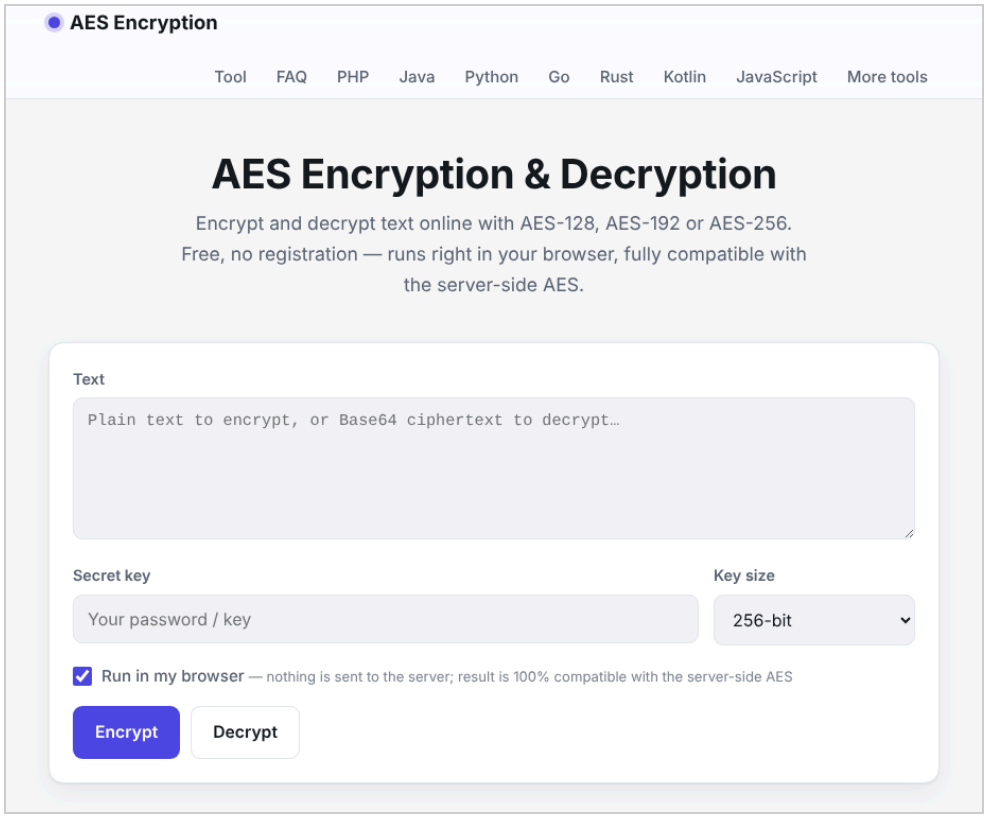
Item	Value
Tool	aesencryption.net
Algorithm	AES, 256-bit key, result in Base64
Pre-shared secret key	`netsec`
Wrong key for the test	`netsec1`

Part 1: Search for an online encoding and decoding tool

I searched for `encrypt decrypt AES online`. The search gives a lot of tools, for example onlineminitools.com, encrypt-online.com and aesencryption.net. They all work the same way: you enter text and a secret key and get the ciphertext back. I chose aesencryption.net, the same tool as in the lab example. It supports 128, 192 and 256-bit keys and shows the result in Base64.



Search results for "encrypt decrypt AES online".



aesencryption.net: text box, secret key, key size and the Encrypt / Decrypt buttons.

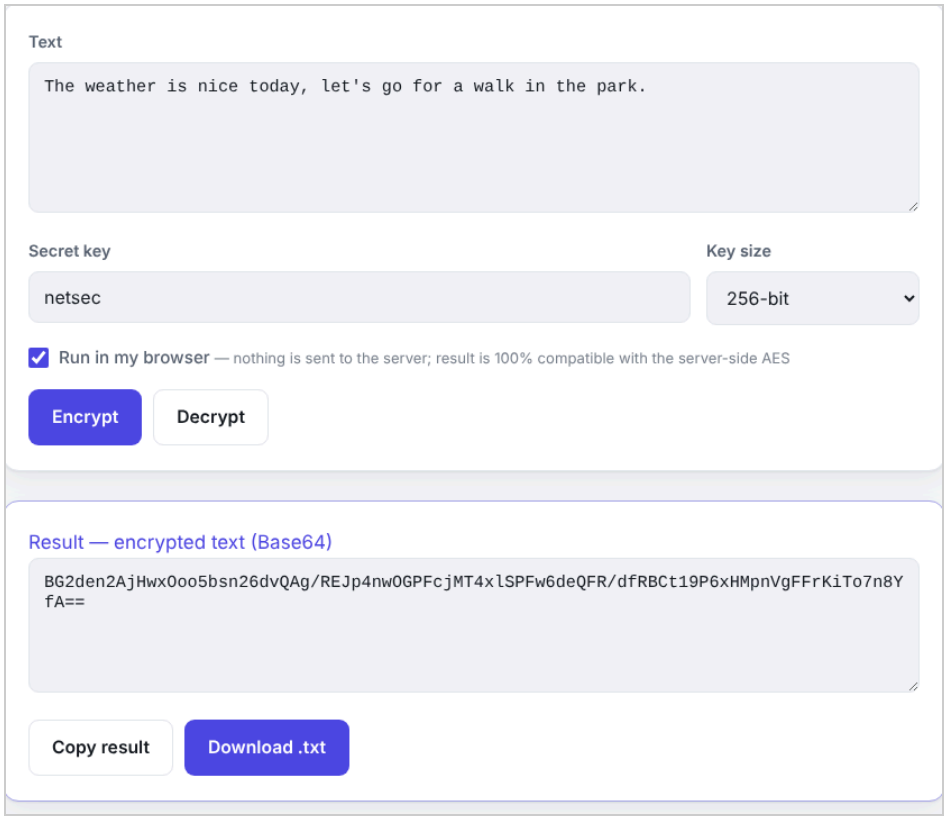
Part 2: Encrypt a message and send it to the lab partner

I entered my plaintext message in the Text box, typed the secret key `netsec`, left the key size at 256-bit and clicked Encrypt.

> `The weather is nice today, let's go for a walk in the park.`

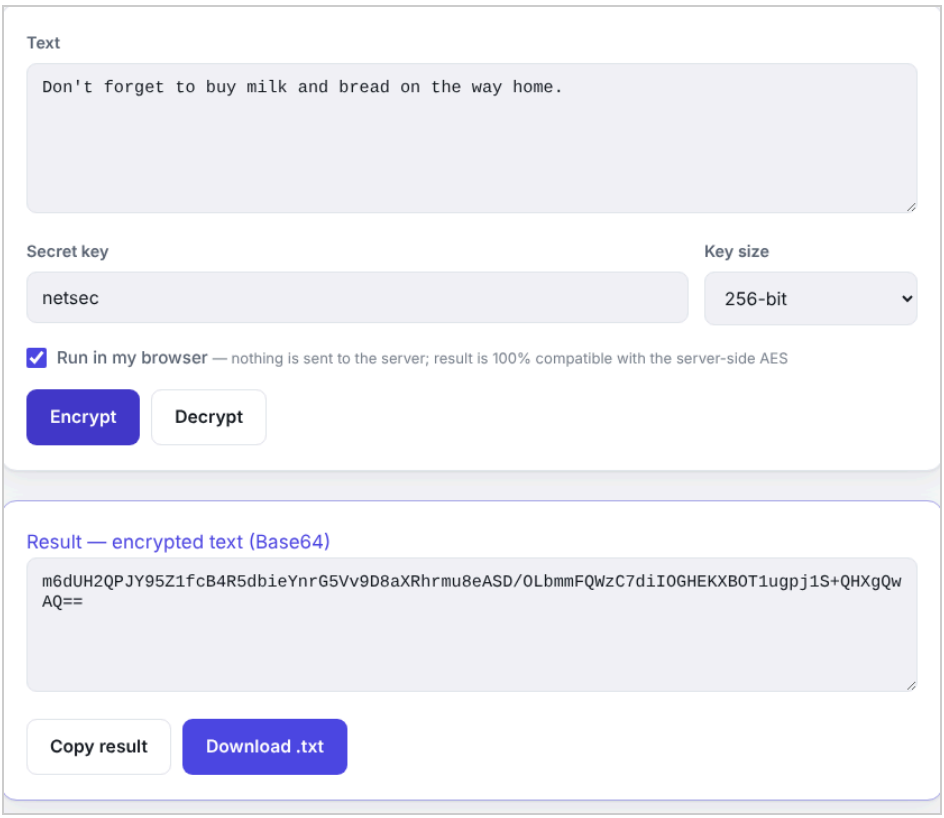
In the result window the message turned into random-looking Base64 text. This is the ciphertext. I copied it and sent it to my lab partner. From the ciphertext alone it is not possible to tell anything about the original message.

> `BG2den2AjHwxOoo5bsn26dvQAg/REJp4nwOGPFcjMT4xlSPFw6deQFR/dfRBCt19P6xHMpnVgFFrKiTo7n8YfA==`



My plaintext encrypted with the key netsec. The result is the ciphertext in Base64.

The partner did the same with their own message and the same key and sent me their ciphertext:



The partner's message encrypted with the same pre-shared key netsec.

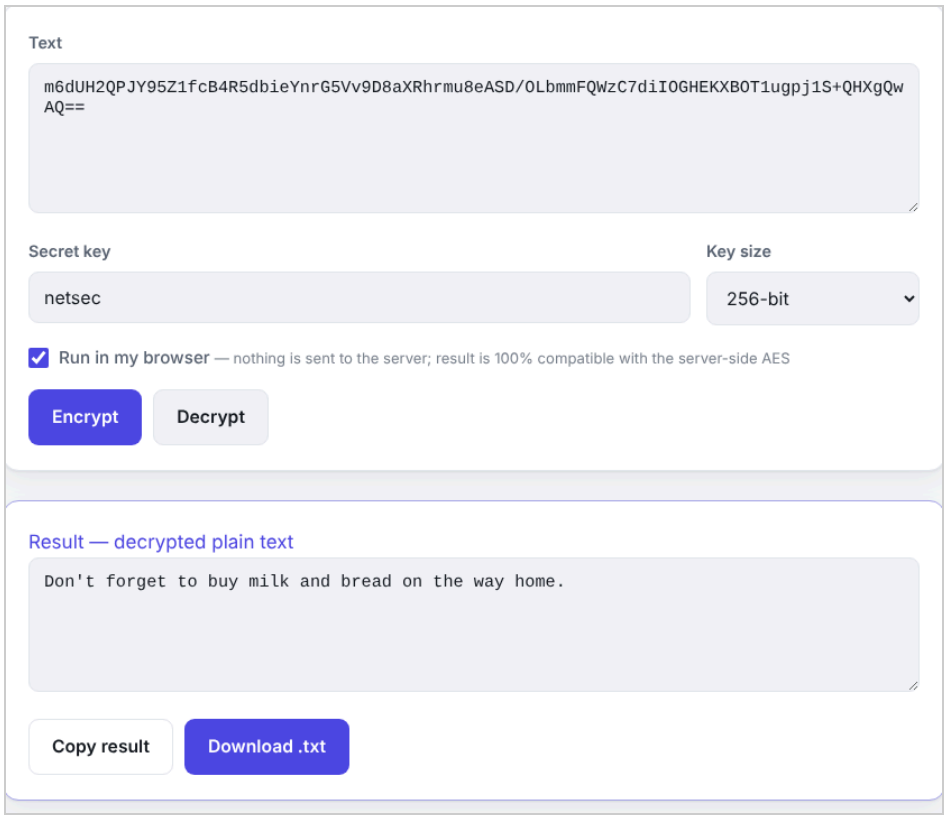
Sender	Plaintext	Ciphertext (Base64)
Me	The weather is nice today, let's go for a walk in the park.	`BG2den2AjHwxOoo5bsn26dvQAg/REJp4nwOGPFcjMT4xlSPFw6deQFR/dfRBCt19P6xHMpnVgFFrKiTo7n8YfA==`
Partner	Don't forget to buy milk and bread on the way home.	`m6dUH2QPJY95Z1fcB4R5dbieYnrG5Vv9D8aXRhrmu8eASD/OLbmmFQWzC7diIOGHEKXBOT1ugpj1S+QHXgQwAQ==`

Part 3: Decrypt the ciphertext

I pasted the ciphertext I received into the Text box, entered the pre-shared key `netsec` and clicked Decrypt. The original cleartext message was displayed.

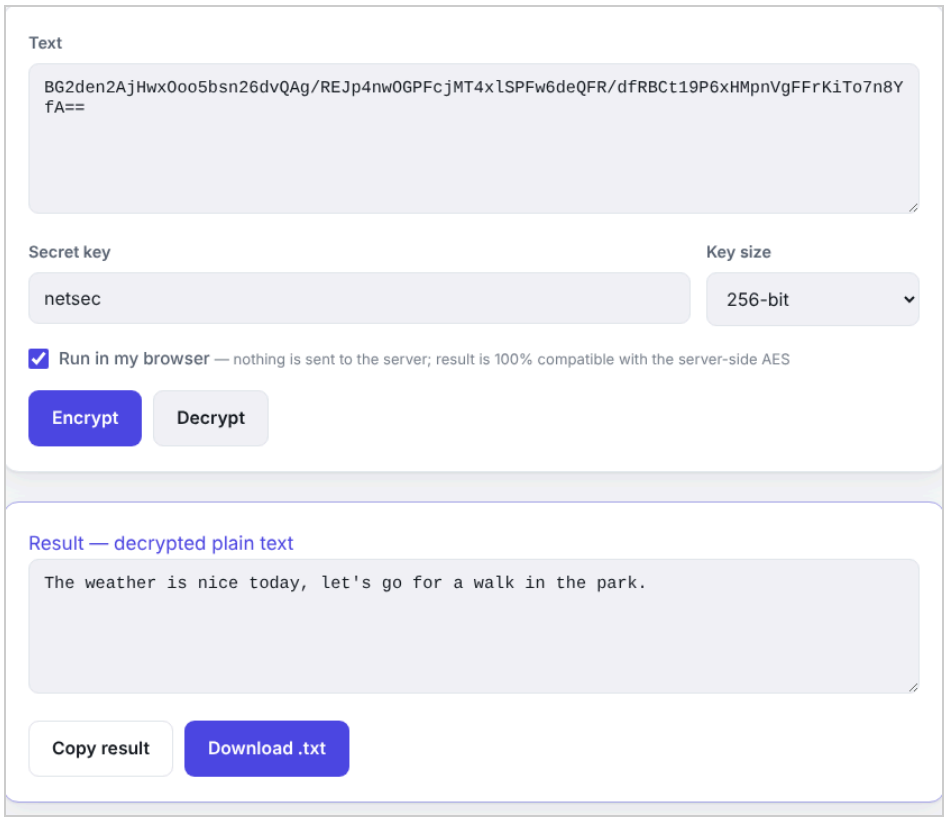
```
m6dUH2QPJY95Z1fcB4R5dbieYnrG5Vv9D8aXRhrmu8eASD/OLbmmFQWzC7diIOGHEKXBOT1ugpj1S+QHXgQwAQ==
```

```
Don't forget to buy milk and bread on the way home.
```



The received ciphertext decrypted with the key netsec gives the original message.

The partner did the same with my ciphertext and got my original message back:

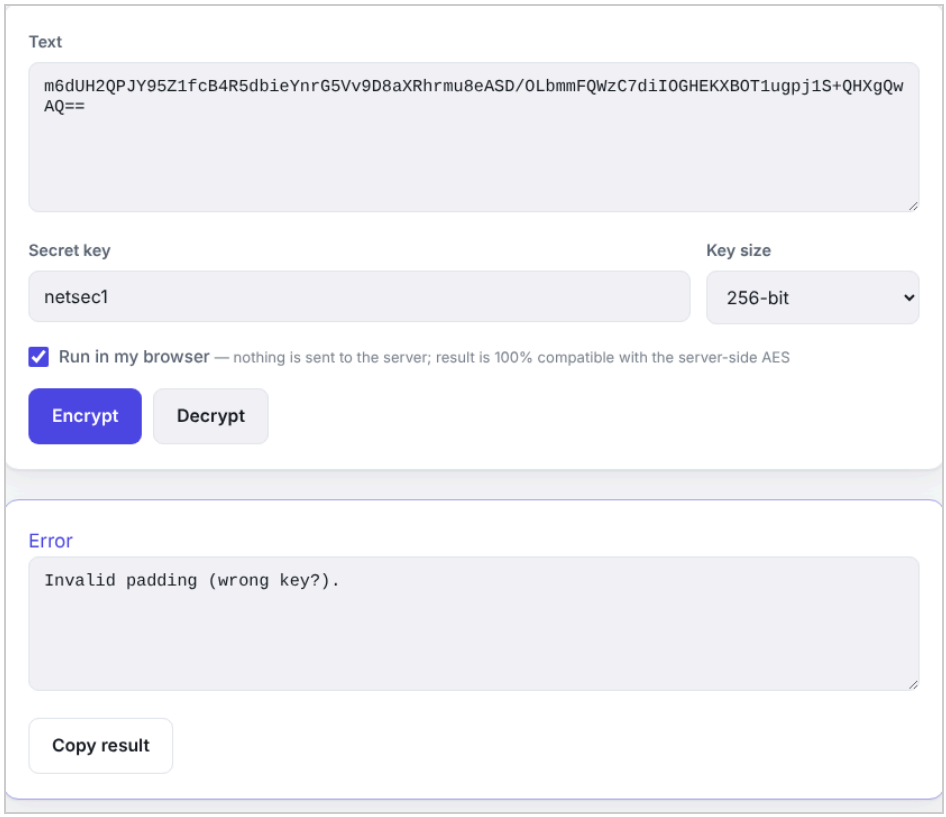


My ciphertext decrypted with the key netsec.

Question

What happens if you use a wrong secret key?

To check it I pasted the same ciphertext and used the key `netsec1`, which is only one character different from the correct one.



Decryption with the wrong key netsec1: the tool shows "Invalid padding (wrong key?)".

The message cannot be decrypted. The tool shows the error "Invalid padding (wrong key?)" and does not show any part of the original text. Even a key that differs by one character gives a completely different AES key, so the result of the decryption is just random bytes. Only someone who knows the exact secret key can read the message. This is also why the key must be shared in a secure way and should be long and hard to guess.

Result

In this lab I learned how symmetric encryption works in practice. With AES the same secret key encrypts and decrypts the message, so the sender and the receiver must agree on the key before they exchange messages. The ciphertext can be sent over an untrusted channel like email, because without the correct key it is useless. The weak point is the key itself: a short key like `netsec` is fine for a lab, but in real life the key should be long and random and must not be sent together with the ciphertext.